

CAPE MAY, NJ, USA

WMO: 745966

Lat: 39.008N

Lon: 74.908W

Elev: 7

StdP: 101.24

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 03726

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.0	(c) -7.2	(d) -18.8	(e) 0.7	(f) -7.9	(g) -16.9	(h) 0.8	(i) -6.1	(j) 11.9	(k) 4.4	(l) 10.7	(m) 3.4	(n) 4.4	(o) 310	(p) 0.496

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	32.3	24.8	31.0	24.0	29.1	23.3	25.7	30.2	25.0	29.1	24.3	28.1	4.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.1	28.0	23.7	18.6	27.5	22.9	17.6	26.7	79.5	30.6	75.9	29.1	73.5	28.2	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.1	(b) 8.8	(c) 8.0	(e) -12.7	(f) 34.1	(g) 2.8	(h) 1.2	(i) -14.7	(j) 34.9	(k) -16.3	(l) 35.6	(m) -17.9	(n) 36.3	(o) -19.9	(p) 37.1		
(a) 10.1	(b) 8.8	(c) 8.0	(e) -13.4	(f) 27.0	(g) 2.9	(h) 1.1	(i) -15.5	(j) 27.7	(k) -17.2	(l) 28.3	(m) -18.8	(n) 28.9	(o) -20.9	(p) 29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.2	1.7	2.8	6.2	11.4	16.4	21.8	24.7	23.7	20.4	14.7	8.8	4.7
	DBStd	9.00	5.40	4.95	4.58	4.39	4.10	3.28	2.77	2.73	3.39	4.28	4.57	4.65
	HDD10.0	896	264	206	135	33	1	0	0	0	0	7	75	175
	HDD18.3	2493	517	434	376	215	89	8	0	1	18	128	286	422
	CDD10.0	2053	5	5	18	74	198	354	454	425	313	153	40	12
	CDD18.3	607	0	0	0	6	28	112	196	167	81	17	1	0
	CDH23.3	4241	0	0	3	41	169	715	1621	1230	406	56	1	0
	CDH26.7	1172	0	0	0	10	37	186	537	326	70	7	0	0
Wind	WSAvg	3.9	4.2	4.0	4.6	4.5	3.9	3.6	3.4	3.2	3.5	3.6	4.0	4.0
Precipitation	PrecAvg	1141	86	73	110	90	81	94	102	125	90	98	79	112
	PrecMax	1505	175	151	231	153	162	256	180	312	178	215	165	223
	PrecMin	844	42	16	10	41	24	38	33	50	10	1	21	44
	PrecStd	138	33	34	53	30	35	57	36	69	45	54	42	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.3	18.0	22.1	27.9	30.2	32.6	34.7	33.3	31.2	27.4	22.0	18.8
		MCWB	13.2	13.5	14.9	18.8	22.3	24.9	26.1	25.7	24.3	21.4	17.0	15.1
	2%	DB	14.0	14.7	17.8	23.7	27.4	30.9	32.5	31.3	28.3	24.8	18.9	16.1
		MCWB	11.6	11.0	12.7	16.5	20.7	23.7	24.8	24.6	22.9	20.6	15.8	13.7
	5%	DB	12.3	12.3	15.3	21.0	25.2	28.7	31.2	29.9	27.2	22.8	17.5	13.9
		MCWB	10.2	9.2	11.1	15.2	19.1	22.8	24.2	24.1	22.1	19.6	14.4	11.8
	10%	DB	9.8	10.0	12.9	17.9	22.8	27.4	29.7	28.2	25.9	21.3	16.2	12.3
		MCWB	7.8	7.5	9.7	13.3	17.9	21.8	23.5	23.1	21.5	18.2	13.3	10.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.1	14.7	16.2	19.8	23.3	25.6	27.2	26.8	25.5	23.2	18.4	15.8
		MCDB	16.1	16.9	20.9	26.2	28.3	31.0	32.3	31.7	28.7	25.4	20.2	18.0
	2%	WB	12.3	11.9	13.7	17.5	21.6	24.5	26.0	25.6	24.3	21.7	16.7	14.1
		MCDB	13.9	13.5	16.9	22.1	26.3	29.2	30.7	29.8	27.3	23.9	18.4	16.0
	5%	WB	10.3	9.5	11.9	15.9	20.2	23.7	25.2	24.9	23.4	20.2	15.1	12.4
		MCDB	12.3	11.6	14.0	19.7	24.4	27.7	29.6	28.6	26.0	22.4	17.2	13.8
	10%	WB	7.8	7.6	10.2	14.3	18.8	22.8	24.4	24.1	22.6	18.8	13.8	10.4
		MCDB	9.3	9.4	13.0	17.6	22.3	26.2	28.3	27.5	24.9	20.8	16.0	12.2
Mean Daily Temperature Range	5% DB	MDBR	7.4	7.9	7.9	8.7	8.6	8.0	8.3	8.0	8.2	8.5	8.0	7.6
		MCDBR	9.8	10.9	11.4	12.5	11.2	9.6	9.5	8.9	8.7	9.1	9.1	9.1
		MCWBR	8.4	8.5	7.9	7.2	6.2	4.9	4.3	4.1	4.7	5.9	7.1	8.3
	5% WB	MCDBR	9.4	10.0	10.0	10.6	10.2	8.9	8.6	7.7	7.3	7.4	7.9	8.7
		MCWBR	8.8	8.8	8.2	7.1	6.2	4.9	4.2	3.9	4.4	5.3	7.2	8.4
Clear-Sky Solar Irradiance	taub	0.317	0.320	0.361	0.391	0.438	0.491	0.504	0.499	0.417	0.363	0.331	0.317	
	taud	2.514	2.507	2.424	2.364	2.263	2.159	2.136	2.134	2.346	2.498	2.536	2.550	
	Ebn at Noon	860	907	897	887	846	800	784	777	827	846	836	834	
	Edh at Noon	77	88	106	120	135	150	152	149	114	88	75	70	
All-Sky Solar Radiation	RadAvg	2.05	2.92	3.94	5.06	5.69	6.22	6.18	5.48	4.49	3.29	2.34	1.80	
	RadStd	0.18	0.31	0.32	0.38	0.65	0.50	0.30	0.37	0.44	0.32	0.21	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+134	+78

Nomenclature: See separate page